

Project 1 - What Can A Starship Do?

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Prompt: Purdue has a fleet of ~60 running starship robots which are currently used to deliver food ordered at campus dining services to hungry boilermakers. Starship delivery robots are present on hundreds of college and corporate campuses across the US and EU. But they don't all deliver food! What else can a starship do?

Goal: In this project, we will explore boilermaker attitudes toward starship robots, automated delivery, and design a unique starship delivery system that goes beyond food service and produces new kinds of human-human connections. Possible design spaces include, but are not limited to, supply and manufacturing delivery, message delivery, lost and found, tours, navigation, or even campus-wide games

1.1 Background Research:

Based on the article Starship Technologies, they conducted an online survey of 7,063 college students on more than 20 campuses across the US where it provides delivery services. The key findings from the article are:

- 6 out of 10 said they could study more because the robots deliver when and where they need them.
- 1 in 4 people say the robots have improved their mental health.
- Nearly two-thirds said they didn't skip meals because they didn't have to wait in long lines for food.
- 30% said they felt safer on campus because of the contactless delivery offered by the robots.
- More than 9 out of 10 say the robots are convenient/helpful.
- More than 40% say they have treated the robot as they would a person by saying thank you, excuse me, etc.
- More than half say they have helped a robot if it appeared to be in need.
- Three-quarters say the robots are friendly/cute.
- The robots are overwhelmingly popular with more than 98% saying they love them or like them on their campus
- More than 90% say they will continue to use autonomous robot delivery if it is available after they graduate.

Based on the academic research on Human-Robot Service Encounters, it shows the progression of findings across three other studies:

- A framework for examining the role of robotics in customers' experiences based on robot visibility, competency and prominence balanced against experiential values.
- Innovation tendencies to play a vital role in whether customers will pay more to coproduce or work together on interactive task with the robots

- Humans follow up on interactive service tasks completed by a robot can enhance customers' satisfaction and perception of value with the experience.

Current Issues with food delivery:

- Slow speeds as they travel at slow walking speed, which can be frustrating for people in hurry especially during peak times
- Limited scope of use which currently focused mainly on just food delivery from restaurants. There is more potential of use beyond that
- Are they environmentally friendly with the use of energy?
- Limited payload and size that they can carry
- Food spills in the robot especially drinks can bounce out the top.

Tertiary Research:

Detecting bio information:

- Air quality
- Temperature
- Wind
- Humidity

1.2 Narrative Methods:

- Data Collection (semi-structured interview)
- Narrative Analysis

1.3 Observational Methods

- How do people interact with robots on sidewalks?
- Shadow a robot (follow one as it makes a delivery)?
- How do robots handle obstacles (curb, stairs, uneven sidewalks)?
- Do robots ever cluster together or what happens when two of them meet at a crossing?

2.1 Interviews Plan

We can all gather 1-2 individuals per person that we will interview. In order to decrease chatter and any bias from previously knowing who we are interviewing, we could interview for each other's chosen peers. As an attempt to reduce the bias in using different interviewers, we will all prepare our questions beforehand and develop a protocol to keep it as consistent as possible. We intend to use the same introduction and explanation of our interview to interviewees as well as asking the same main and probing questions and wrapping it up the same. We will also make sure to audio record our interviews and keep the pace moving as consistently as possible. We can then all come together and listen to these recordings and share our findings such that all interviewees/ team members are on the same page and understanding of our interview results.

Below is our first attempt at the first round of interviews:

Questions:

- Have you used a starship before? Do you regularly use starships?
If so, how often? If not, when did you use them regularly?
- What are your frustrations and/or enjoyments when using them? Can you tell me a story about a time you used a starship? What's the funniest thing you've ever seen them do?
- If you saw a starship get hit by a car, what would your initial reaction be? Would you laugh? Feel bad for it? Help it?
- Would you be more likely to use a starship if there was a higher level of customization? (ex: adding a clip of your voice or a saying for ai to play when starship arrives, or adding an image to display when it arrives?)
- What are your thoughts on the format of a starship - that it delivers something to you from different places on campus? Do you find this format useful/convenient?
- Would you like if starships were used for purposes not related to delivery? (ex: making its movement or exterior the purpose, rather than what can be held inside)
- Would the frustrations you mentioned earlier still be relevant if the starships weren't used for delivery?

Findings:

2.2 Shayla's Interview

- Frustrations: takes too long
- She ordered food from a starship in a snow storm so she didn't have to go out. The starship got stuck so she had to go out to get the food anyways
- Saw one get hit by a car
- They're helpful for people who are injured who can't walk to get food
- Would think it was funny if there was more customization
- Thinks they could be used for purposes not related to delivery, but says the design would need to be changed: make taller, larger screen, turn better
- Instead of making them move on a constant route or sit in front of buildings, make it so that you could bring one to yourself
- Make more interactive with touch screen
- Frustrations from earlier would be resolved with this design

2.4 Syahirah's Interview

Interviewee background: A sophomore lives in a residence hall, has experienced using Starships

Key findings:

- The interviewee uses Starships occasionally and the usage tends to increase in bad weather or when convenience outweighs the effort of walking around the campus
- Starships can be slow or easily stuck on curbs and obstacles
- Willingness to assist if needed, but also scared it may be worse
- Has a strong interest in customization to make the experience more entertaining like automated voices
- Current design is adequate for food delivery but could be improved for other purposes like organization's newsletter or message delivery
- Adding features like screen or facial features to give starships more personality but cost and practicality may limit adoption

UPDATED FIRST ROUND INTERVIEW:

Intro: Hello, my name is Kayla and I just wanted to thank you for taking the time to complete this interview with me today. I want to let you know that all of your information will be confidential and anonymous. Before we get into things I also want to confirm that it is okay that this interview is audio recorded for the intent of gathering notes and reviewing accurate information given from you after. If at any point you wish to leave the interview you may and feel free to ask any questions along the process. To brief you on the topic at hand before we jump right into it, today my team and I are hoping to understand your experience with the mini robots on campus commonly called Starships.

Questions:

1. Think back to the last time you or someone you know used a Starship delivery. Could you describe the experience from the moment you placed the order until you opened the lid?
 - a. What did you order and why? Do you remember how long it took?
2. What are your frustrations and enjoyments of Starships?
 - a. Do you think the majority of campus agrees or disagrees?
3. If you were given the task to fix a problem with the current Starship experience, what would you work on first?
 - a. Please explain your choice and why it's the first thing that needs improved.
 - b. What functions (if any) would you keep the same?
4. Do you believe the future of Starships lies in simply making their delivery skills better, or should they be used for a totally different purpose?
 - a. If you chose the latter, what one function would you assign the robot, and how would it need to change physically (size, shape, features) to succeed at that new job?

Conclusion: That wraps up the questions I have today and I want to thank you for your time and valuable feedback. We are conducting these interviews in hopes of improving the user experience of the on campus Starships and exploring their potential. If you have any questions, please feel free to ask and if you need to reach out to me for any reason you can contact me via email.

Findings:

- Alyssa:

<https://drive.google.com/file/d/111la5QaR7VjnEGnH7xJFyhYptFrbDZUo/view?usp=sharing>

has trouble navigating, doesn't compete with top competitors (DoorDash), takes a lot of time, could see use in it for a different purpose

Amanda 1:

<https://drive.google.com/file/d/1YxemgbHtM5t4lxVlv4kzyylhi-jN-pc/view?usp=sharing>

- inaccurate refilling, sometimes gross, "annoying"
- Not a quick refilling process
- Blocks sidewalks, inconvenient, likes how they mingle at midnight, neat to watch
- Updating sensors, more interactive (Stu's can click a button)
- Would keep how they look
- Should have a different purpose
- Size could be changes depending on its use

Lillia round 1:

https://drive.google.com/file/d/1R4enkTBDXz4rWresz8j8uJf6blu_LaPh/view?usp=sharing

- frustrating wait, 30-45 mins
- Cute/ cool experience, one time thing
- No regular users
- Most people would rather walk m, beneficial to her friend with broken leg
- Change up routes to avoid areas of struggle like crosswalks/ busy stop signs
- Open to see other options to utilize starships
- Could have students bike and do delivery versus subpar mini robots— they have a better purpose like navigating small spaces and trying to track patterns on campus

SECOND ROUND INTERVIEW QUESTIONS:

Questions:

1. If the Starship were more of a mobile campus resource than a delivery vehicle, how would that change your relationship with it?
2. If you saw a Starship, what is the single piece of real-time information that would make you stop what you were doing to check its screen? Now, what piece of information would make you completely ignore it?

3. Describe the *ideal* spot for this information resource. Conversely, where on campus would its presence potentially annoy you or feel like an unwelcome obstruction?
4. Since this robot's main job would switch to display, imagine its entire exterior casing is a screen. If you could physically touch up the robot, what design switches if any, would you suggest?
5. What do you think we could do with Starships to make digesting the information from these mini robots more unique than simply finding it on your phone? Would you find benefit with them or be more likely to continue looking at your phone for this information instead?

Findings:

- Audio:
- Notes:

ANOT round 2:

https://drive.google.com/file/d/1SurXswZopVcFOCxWN8Goz0K_H9khyMoj/view?usp=sharing

- more usage depending on resource
- Date + time would be good to include, ads selling things annoying
- Ideal spot: near WALC, STEW, PMU to avoid annoyance of sidewalks getting clustered. Those buildings since they would be good for off campus students too (central locations)
- In design switch: color theory/ update with seasons, make interactive, use for surveys, more approachable/ appealing
- Make it like a game to engage and stop people for a couple minutes

Lillia round 2:

<https://drive.google.com/file/d/13gkUzy15eh2pK-xA3nLnPT3zINQ32Zuh/view?usp=sharing>

- would be more interactive for off campus students too
- Good info: more decorative, make use of the flags
- Ignoring info: lots of text
- Ideal spot: memorial mall, bus stops

- Reminders of emails / newsletters for different schools

3.1 Ideas

Brainstorming:

1. Delivering stuffs from friends-to-friends, organizations
2. Mobile charging station
3. First aid delivery
4. Mobile bulletin board
 - a. Have a screen to display news, promotion, etc.

5. Data collection

- a. Sidewalk/ pathway congestion
- b. Noise pollution data (construction, traffic, student activity)
- c. Building busyness
- d. Environmental monitoring:
 - i. Weather quality: tracking CO2, pollution, temperature, humidity
 - ii. Bio diversity monitoring: animal presence, soil quality, plant growth, invasive species

6. Have built-in voice activation & communication

- a. Polite social phrases
 - i. "Excuse me"
 - ii. "Thank you"
- b. Safety & awareness phrases
 - i. "Watch out, behind you"
- c. Accessibility phrases
 - i. "Help"
- d. Personalized messages
 - i. "Happy Birthday"

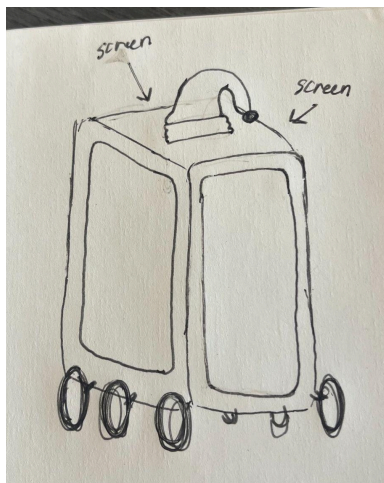
Final Idea:

Taller robot with 4 screens

- Screen 1: Weather
- Screen 2: Encouraging Quotes
- Screen 3: Events
- Screen 4: Dining Hall Capacity

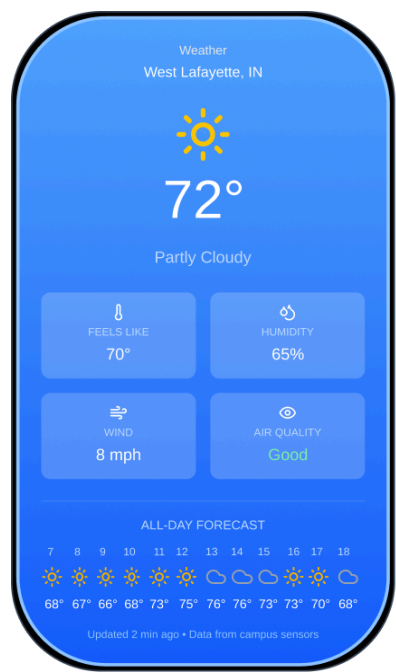
4.1 Prototyping

Sketches



Wireframes

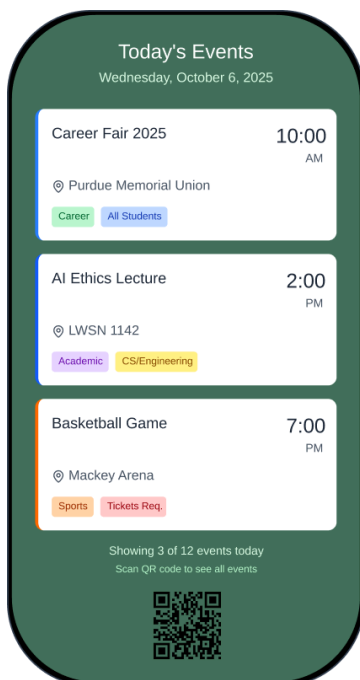
1. Screen 1: Weather Forecast



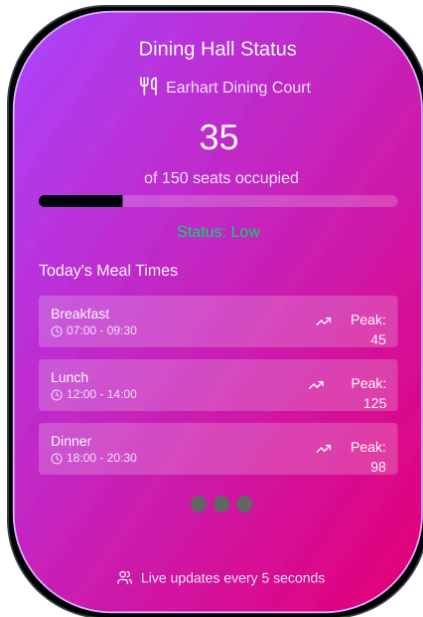
2. Screen 2: Inspirational Quotes



3. Screen 3: Campus Events



4. Screen 4: Dining Hall Capacity



5.1 Design Evaluation

- Very fast when people aren't out (during non busy times on campus/between classes)
- Doesn't care, almost ran into me
- Grouped together on corner and were overwhelmed by each other
- Clustering = catches attention
- Ask questions like: what's the funniest thing you've ever seen them do?

References

Hoffa, R. (2023, May 16). *Exploring consumer perceptions of human-robot service encounters*. News | College of Health and Human Sciences.

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National study of more than 7,000 college students shows delivery robots help students feel safer, skip fewer meals and improve their mental health. (2024, November 13). Starship Technologies: Autonomous robot delivery - The future of delivery - today!.

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